

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 90%. We keep your highest score.

Next item →

1.

In PCA, the variance explained by each principal component is quantified by its 

Eigen

 value.

1 / 1 point
- Response 1

☒ Eigen

☐ Covariance

☐ Singular

☐ Principal

☒ Nice work

2.

The matrix decomposition technique that expresses a matrix as the product of an orthogonal matrix, a diagonal matrix, and another orthogonal matrix is called 

Singular

 Value Decomposition.

1 / 1 point

Response 1

☐ Covariance

☐ Eigen

☒ Singular

☐ Principal

☒ Nice work

3.

he eigenvectors of the covariance matrix in PCA are called 

Principal

 components.

1 / 1 point

Response 1

☐ Singular

☒ Principal

☐ Covariance

☐ Eigen

☒ Nice work

4.

PCA projects data onto a new coordinate system such that the greatest variance lies on the first axis, known as the 

Principal

 component.

1 / 1 point

Response 1

☐ Covariance

☐ Eigen

☒ Principal

☐ Singular

☒ Nice work

5.

Select if the following statements are true or false.

5 / 5 points

True

Principal Component Analysis (PCA) is used to reduce the dimensionality of a dataset while preserving as much variance as possible.

True

Singular Value Decomposition (SVD) decomposes a matrix into the product of three matrices: a diagonal matrix, and two orthogonal matrices.

False

PCA can only be applied to datasets with numeric features.

True

In PCA, the principal components are always orthogonal to each other.

False

Singular Value Decomposition is useful for solving linear regression problems.

Response 1

☒ True

☐ False

☒ Nice work

Response 2

☒ True

☐ False

☒ Nice work

Response 3

☐ True

☒ False

☒ Nice work

PCA can be applied to numeric datasets, but preprocessing is required for categorical data.

Response 4

☒ True

☐ False

☒ Nice work

Response 5

☐ True

☒ False

☒ Nice work

SVD is primarily used for matrix factorization, but it can be applied in various contexts including linear regression.

6.

The matrix used in SVD that contains the right singular vectors is denoted as 

V

.

1 / 1 point

Response 1

☐ U

☐ Z (Sigma)

☒ V

☐ A

☒ Nice work

7.

Which of the following is not a step in PCA?

1 / 1 point

☐ Standardizing the data

☒ Performing k-means clustering

☐ Computing the covariance matrix

☐ Calculating eigenvalues and eigenvectors

☒ Nice work

8.

In SVD, which matrix contains the singular values?

1 / 1 point

☐ The matrix Z (Sigma)

☐ The matrix A

☐ The matrix U

☐ The matrix V

☒ Nice work

9.

What is the primary goal of applying PCA?

1 / 1 point

☐ To identify outliers

☐ To compute the inverse matrix

☒ To reduce the number of features while retaining the most variance

☐ To perform classification

☒ Nice work

10.

Which of the following statements about SVD is true?

1 / 1 point

☐ SVD is used exclusively for dimensionality reduction

☐ SVD decomposes a matrix into two diagonal matrices

☐ SVD requires the matrix to be square

☒ SVD can be applied to any rectangular matrix

☒ Nice work

11.

Which method is used to ensure that principal components are uncorrelated?

1 / 1 point

☐ Singular Value Decomposition

☐ Principal Component Analysis

☐ Feature scaling

☒ Orthogonalization

☒ Nice work